

Aggregation Semantics for Prioritization in Cooperative Automated Decision Making

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Abstract

To be rewritten.

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1. Introduction

1.1. Research Questions and Contributions

2. Background and Related Work

2.1. Context-Aware and Cooperative Decision Support Systems

The increasing adoption of Artificial Intelligence (AI) in decision-support systems has highlighted the need to complement predictive performance with contextual, organizational, and domain-specific knowledge. In many real-world settings, decisions cannot be based solely on statistical predictions because institutional policies, expert knowledge, regulatory requirements, and operational constraints must also be considered. As a result, a growing body of research has focused on context-aware and cooperative decision-support systems capable of integrating heterogeneous sources of information into the decision-making process.

One important research direction concerns the incorporation of explicit knowledge representations alongside data-driven models. Early examples include cognitive-agent architectures that combine symbolic rules with machine-learning mechanisms to support context-aware decision making [1]. Similarly, ontology-based clinical decision-support systems have employed contextual ontologies and clinical pathways to select context-appropriate recommendations for individual cases [2]. More recent developments have extended these ideas through semantic models, knowledge graphs, and hybrid AI architectures that combine predictive models with structured domain knowledge to improve decision quality and interpretability [3,4].

A second research stream emphasizes cooperation between human and artificial intelligence. Rather than viewing decision support as a purely predictive task, these approaches consider decision making as a collaborative process in which machine-learning models provide evidence while human experts contribute contextual understanding, oversight, and governance. Cronholm and Göbel [5], for example, argue that effective decision support requires appropriate mechanisms for combining human and artificial intelligence rather than simply improving predictive accuracy. Likewise, recent surveys on human-centred AI highlight transparency, accountability, contextual awareness, and governance as fundamental requirements for trustworthy decision-support systems [6].

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Related ideas also appear in cooperative and distributed decision-making environments. Multi-agent systems, collective-intelligence frameworks, and cooperative AI architectures frequently rely on multiple decision sources that must be reconciled before a final recommendation is produced [7]. In these settings, contextual information often plays a coordinating role by constraining, guiding, or validating the outputs generated by predictive components. Similar concerns arise in governance-oriented decision-support systems, where contextual information may encode operational procedures, organizational priorities, safety requirements, or regulatory obligations [4,8].

Taken together, these studies clearly demonstrate the importance of contextual knowledge in decision-support systems. However, their primary focus typically lies in knowledge representation, ontology design, human–AI interaction, governance mechanisms, or system architecture. Once predictive evidence and contextual assessments have been generated, the numerical integration process that combines both sources of information is often treated as a secondary implementation detail. Consequently, relatively little attention has been devoted to understanding how alternative integration mechanisms affect the final recommendations produced by the system.

This limitation becomes particularly visible in architectures that explicitly separate predictive evidence from contextual assessment. Recent work on Cooperative Automated Decision-Making Systems (CADEMAS) [9] and its machine-learning instantiation CADEMAs-ML [10] exemplifies this design philosophy. CADEMAs organizes heterogeneous automated decision-making components and contextual knowledge into distinct layers, allowing predictive evidence and contextual assessments to be generated independently before being combined into a final recommendation. Such architectures make the integration mechanism itself an explicit design choice rather than an implicit implementation detail. As a result, the aggregation operator determines how predictive evidence interacts with contextual requirements, organizational priorities, or policy constraints, thereby shaping the semantics of the resulting decision process.

This observation suggests that aggregation should not be viewed merely as a numerical post-processing step. Instead, it can be interpreted as a governance mechanism that regulates the interaction between predictive and contextual information. Understanding the implications of different aggregation semantics therefore becomes essential for the design of cooperative and context-aware decision-support systems. The next subsection reviews the aggregation literature from this perspective.

2.2. Aggregation Operators and Decision Semantics

Aggregation operators constitute one of the central building blocks of multi-criteria decision making (MCDM), fuzzy decision support, and collective decision-making systems. Their purpose is to combine multiple sources of information into a single assessment while preserving, to a greater or lesser extent, the decision semantics associated with the original criteria. Consequently, aggregation has been studied extensively from both theoretical and applied perspectives.

Among the most influential aggregation frameworks are Ordered Weighted Averaging (OWA) operators [11,12], t-norm and t-conorm based operators, geometric aggregation functions, and non-additive integrals such as the Choquet and Sugeno integrals. Collectively, these approaches provide a rich mathematical vocabulary for representing different attitudes toward compensation, conjunction, disjunction, and interaction among criteria. From this perspective, the choice of aggregation operator is not merely a numerical convenience but a modeling decision that determines how favorable and unfavorable evidence may compensate for one another.

A substantial body of literature has explored these properties in traditional decision-making settings. Reviews by Mardani et al. [13] and Csiszár [14] highlight the widespread use of OWA-based aggregation and related fuzzy operators across diverse application domains. More recent studies have continued to investigate how aggregation semantics influence consensus formation, robustness, criterion interaction, and ranking behavior. Examples include consensus-oriented aggregation in group decision making [15,16], robust Choquet-based decision models [17,18], and applications employing geometric and OWA-type operators to capture alternative decision attitudes [19,20].

Despite this methodological richness, most aggregation studies are formulated within classical MCDM settings, where the aggregated inputs represent multiple criteria, attributes, preferences, or expert judgments associated with the same decision problem. The primary objective is typically to identify the most appropriate operator for combining these criteria or to develop new aggregation mechanisms with desirable mathematical properties. Comparatively less attention has been paid to architectures in which the aggregated inputs originate from fundamentally different decision layers.

This distinction is particularly important in cooperative automated decision-making systems. In architectures such as CADEMAs-ML, the aggregation process does not combine multiple criteria describing an alternative. Instead, it reconciles two conceptually distinct sources of evidence: a predictive assessment generated by machine-learning models and a contextual assessment encoding organizational priorities, regulatory requirements, expert knowledge, or policy constraints. Under this interpretation, the aggregation operator effectively determines the extent to which predictive evidence may compensate for weak contextual support, or conversely, whether contextual constraints can restrict otherwise favorable predictions. The operator therefore acts as an implicit governance mechanism rather than merely a criterion-combination tool.

For this reason, the present work focuses on three representative aggregation regimes exhibiting fundamentally different compensation semantics. The weighted arithmetic mean represents a fully compensatory integration strategy in which predictive and contextual information may offset one another. The weighted geometric mean represents a partially compensatory strategy that penalizes strong imbalances between both sources of evidence. Finally, the minimum operator represents a non-compensatory strategy in which the weakest component determines the final assessment. Together, these operators span a spectrum ranging from highly permissive to highly restrictive integration behavior while remaining interpretable and widely understood within the aggregation literature.

From the perspective of this study, the objective is therefore not to propose a new aggregation operator nor to extend aggregation theory. Instead, aggregation operators are treated as alternative decision semantics within a cooperative automated decision-making architecture. The key question is how different compensation regimes influence policy compliance, prioritization behavior, robustness to predictive perturbations, and ranking stability when predictive and contextual information must jointly determine the final recommendation.

2.3. Research Gap and Positioning

The reviewed literature reveals two largely independent research streams. On the one hand, context-aware and cooperative decision-support systems have developed increasingly sophisticated mechanisms for incorporating contextual knowledge into predictive processes through ontologies, symbolic rules, knowledge graphs, human oversight, and governance-oriented architectures [1,2,5,6]. These studies demonstrate the importance of contextual information but typically focus on knowledge representation, rule selection, system architecture, or human–AI interaction. Once predictive and contextual assessments

have been generated, the mechanism responsible for their numerical integration is often treated as an implementation detail.

On the other hand, aggregation theory and fuzzy MCDM have developed a rich collection of operators with diverse compensation semantics, including OWA operators, geometric means, t-norms, Choquet integrals, and Sugeno integrals [11–14]. This literature provides extensive theoretical knowledge regarding compensation, criterion interaction, robustness, consensus formation, and preference aggregation. However, aggregation operators are generally studied as mathematical constructs or as components of multicriteria decision methods, where the aggregated inputs represent criteria, preferences, or expert judgments associated with the same decision problem.

Consequently, a gap exists between context-aware decision-support frameworks and aggregation theory. The former provides mechanisms for representing contextual knowledge but rarely investigates how different aggregation semantics affect the preservation of contextual constraints after integration. The latter offers powerful aggregation mechanisms but seldom studies their implications when heterogeneous decision signals, such as predictive machine-learning outputs and contextual assessments, must jointly determine operational recommendations.

From this perspective, the gap addressed in this work is not the absence of aggregation operators, nor the lack of mathematical knowledge about their properties. Rather, it is the lack of a systematic understanding of aggregation as a governance mechanism within cooperative automated decision-making architectures. In particular, relatively little attention has been devoted to questions such as whether contextual assessments can effectively act as veto mechanisms, how different compensation regimes influence policy compliance, how sensitive prioritization outcomes are to predictive overconfidence, or how ranking stability is affected by uncertainty in contextual information.

These observations motivate the following research questions:

- *RQ1. How do different aggregation semantics affect the interaction between predictive evidence and contextual assessments in a cooperative automated decision-making architecture?*
- *RQ2. Under what conditions can aggregation operators preserve or violate contextual constraints and veto-like policies?*
- *RQ3. How do alternative aggregation regimes influence prioritization robustness under predictive overconfidence and contextual uncertainty?*
- *RQ4. To what extent do different aggregation semantics alter the stability of the resulting rankings and intervention priorities?*

Unlike traditional fuzzy MCDM studies, the objective of the present work is not to propose a new aggregation operator, derive new algebraic results, or solve a classical multicriteria decision problem. Likewise, unlike many context-aware DSS frameworks, the focus is not on developing new mechanisms for representing contextual knowledge. Instead, aggregation operators are interpreted as alternative policy semantics within the CADEMAs architecture. By keeping both the predictive subsystem and the contextual subsystem fixed, the study isolates the aggregation mechanism as an independent design variable and examines its implications for policy compliance, robustness, and prioritization behavior. This positioning distinguishes the contribution from both aggregation theory and context-aware decision-support research.

2.4. Chronological Comparison of Related Work

Table 1 summarizes representative contributions discussed in the previous subsections and positions the present work with respect to the existing literature. The comparison focuses on four dimensions that emerge from the identified research gap: (i) the explicit integration of predictive and contextual information, (ii) the treatment of contextual con-

Table 1. Chronological comparison of representative related work. “Partial” indicates that a contribution addresses the dimension only indirectly (e.g., through rules, consensus thresholds, or architectural design) without a dedicated analysis of predictive–contextual score fusion.

Reference	Year	Main focus	Predictive + contextual integration	Policy / veto analysis	Aggregation robustness	Cooperative ADM
Teng and Tan [1]	2008	Context-aware cognitive agents	✓	Partial	No	Partial
Yao and Kumar [2]	2012	Ontology-based clinical DSS	✓	Partial	No	No
Merkert et al. [21]	2015	ML in DSS survey	Partial	No	No	No
Moral et al. [15]	2017	Consensus aggregation	No	No	Partial	No
Mardani et al. [13]	2018	Fuzzy MCDM review	No	No	No	No
Csiszár [14]	2021	OWA theory	No	Partial	Partial	No
Cronholm and Göbel [5]	2022	Human-centred AI DSS	✓	Partial	No	Partial
Yang et al. [17]	2022	Robust Choquet-based MCDM	No	Partial	✓	No
Liao et al. [18]	2023	Contextual Choquet models	Partial	No	✓	No
Trillo et al. [16]	2023	Consensus-based GDM	No	Partial	No	No
Fanfani et al. [6]	2025	Human-centred AI survey	✓	Partial	No	Partial
Present work	2026	CADEMAS aggregation semantics	✓	✓	✓	✓

straints or veto-like policies, (iii) the analysis of aggregation robustness, and (iv) the study of aggregation within cooperative automated decision-making architectures.

The table illustrates that prior research has typically focused either on context-aware decision-support architectures or on aggregation methods in MCDM and fuzzy decision making. Comparatively little attention has been devoted to aggregation as a governance mechanism that simultaneously integrates predictive evidence, contextual knowledge, policy constraints, and robustness considerations within a cooperative ADM framework.

3. CADEMÁS and CADEMÁS-ML

The Cooperative Automated Decision-Making System (CADEMÁS) framework was introduced by Novoa-Hernández et al. [9] as a general architecture for context-aware decision support in which multiple heterogeneous Automated Decision-Making (ADM) units cooperate to produce a unified recommendation, score, or decision. Rather than prescribing a specific decision model, CADEMÁS provides a modular structure that separates the generation of decision evidence from its contextual evaluation and final integration.

This separation is particularly relevant for the problem studied in the present work. Many decision-support systems combine predictive evidence with contextual information, but these elements are often intertwined within the same model or decision process. In contrast, CADEMÁS explicitly distinguishes three conceptually different functions: (i)

evidence generation, (ii) contextual assessment, and (iii) decision integration. Consequently, the aggregation mechanism can be analysed independently of both the predictive subsystem and the contextual knowledge representation.

Formally, a CADEMAs instance is represented by the tuple

$$\mathcal{S} = \langle I, U, O, K, T \rangle, \quad (1)$$

where I denotes the Input Manager, U the set of ADM units, O the Decision Integration Layer, K the Contextual Modulator, and T the universe of admissible data representations.

Given an input case, the ADM units independently generate decision evidence from the available information. In parallel, the Contextual Modulator evaluates the extent to which the same case satisfies organizational priorities, expert knowledge, regulatory requirements, or other contextual constraints. The outputs of both subsystems are then combined by the Decision Integration Layer to produce the final decision score.

An important specialization of this framework is CADEMAs-ML [10], in which the ADM units correspond to machine-learning models. In this setting, multiple predictive models cooperate to produce a consolidated predictive assessment, while contextual information is represented through a fuzzy inference system inspired by Pérez-Cañedo et al. [22] and Pérez-Cañedo et al. [23]. The contextual subsystem translates organizational priorities and institutional requirements into a normalized degree of contextual satisfaction.

To focus on the role of aggregation, the internal implementation of both subsystems is abstracted in terms of two normalized scores. For each decision case i , the predictive subsystem produces a predictive score

$$R_i \in [0, 1], \quad (2)$$

which summarizes the cooperative evidence generated by the predictive ADM units. Depending on the application domain, R_i may represent a risk estimate, a confidence score, a probability of occurrence, or any other normalized predictive assessment.

Independently, the contextual subsystem produces a context score

$$Q_i \in [0, 1], \quad (3)$$

which quantifies the degree to which the case satisfies the contextual requirements encoded by the organization, institution, or application domain.

It is worth noting that the restriction of both predictive and contextual scores to the unit interval does not limit the generality of the proposed formulation. Any continuous score defined over an interval $[a, b]$ can be transformed into the normalized domain $[0, 1]$ through an affine mapping. This transformation preserves ordering relations and therefore maintains the ranking semantics required in prioritization tasks. Consequently, R_i and Q_i should be interpreted as normalized semantic scores rather than measurements expressed in a physical unit. The use of the unit interval allows the analysis of aggregation properties independently of the original scale of the predictive and contextual information. More importantly, the lower boundary $Q_i = 0$ represents a complete *contextual violation*.

The final prioritization score is then obtained through an aggregation operator

$$P_i = A(R_i, Q_i), \quad (4)$$

where A defines the integration semantics between predictive evidence and contextual assessment. Candidate cases are subsequently ranked according to P_i , and the corresponding Top- K subset is selected for intervention or resource allocation.

It is important to emphasize that the aggregation problem considered here differs from traditional fuzzy MCDM settings. In classical MCDM, aggregation operators combine multiple criteria describing different attributes of the same alternative, typically with the objective of capturing trade-offs among decision criteria. In CADEMAs-ML, however, the operator combines heterogeneous sources of decision evidence that play fundamentally different roles: a predictive assessment generated by cooperating machine-learning models and a contextual assessment derived from organizational, regulatory, or domain-specific knowledge.

Consequently, the aggregation operator does not merely determine the relative importance of criteria. Instead, it governs how predictive evidence interacts with contextual requirements and therefore acts as an explicit policy-design component within the decision architecture. The central question is not which operator produces the most desirable aggregate score in a multicriteria sense, but rather how alternative aggregation semantics affect policy compliance, contextual veto preservation, robustness to predictive overconfidence, and ranking stability under uncertainty.

Existing CADEMAs-ML applications employ a weighted linear combination of predictive and contextual scores [10]. However, the CADEMAs framework itself does not prescribe any particular aggregation operator [9]. Consequently, different aggregation semantics can be introduced without modifying either the predictive subsystem or the contextual subsystem. This modularity provides an ideal setting for studying aggregation as a policy-design mechanism and for analysing how different compensation regimes affect prioritization, policy compliance, robustness, and ranking stability. The following section introduces the aggregation operators considered in this study and develops the theoretical properties that motivate the subsequent experimental analysis.

4. Aggregation Semantics and Theoretical Analysis

The aggregation mechanism in CADEMAs-ML determines how predictive evidence and contextual knowledge are integrated into a unified prioritization decision. Therefore, the aggregation operator should not be interpreted as a mere mathematical component, but as an explicit design choice that defines the interaction between machine-learning evidence and contextual requirements.

This section provides the theoretical foundations required to answer the research questions introduced in Section 2.3, particularly those related to the role of aggregation semantics in cooperative automated decision-making. Specifically, the analysis addresses how different aggregation regimes modify the interaction between predictive and contextual information (RQ1), under which conditions contextual constraints and veto-like policies are preserved or violated (RQ2), how aggregation semantics influence robustness to predictive overconfidence and contextual uncertainty (RQ3), and how they affect the stability of the resulting prioritization rankings (RQ4).

The analysis does not aim to introduce a new aggregation operator or to extend classical aggregation theory. Instead, it studies existing aggregation paradigms as alternative policy semantics within the CADEMAs-ML architecture. The section is organized in three steps. It first formalizes the structural properties that characterize aggregation behavior, including boundedness, monotonicity, continuity, zero absorption, contextual veto preservation, and compensability. It then examines three representative aggregation regimes with different decision semantics: fully compensatory linear aggregation, sub-compensatory aggregation through the weighted geometric mean, and non-compensatory minimum aggregation. It closes by deriving conditions for contextual veto preservation, rank reversal under compensation, and sensitivity to perturbations in predictive and contextual assessments.

These results provide a formal basis for interpreting aggregation operators as policy-design components in cooperative automated decision-making systems and motivate the experimental evaluation of their practical consequences in the following sections.

4.1. Structural Properties of Aggregation Operators

Without loss of generality, all scores are assumed to be normalized within the unit interval. Any bounded continuous scoring scale $[a, b]$ can be transformed into this representation through a monotonic affine transformation, preserving the ordering relations and the aggregation semantics considered in this work.

Let

$$A : [0, 1]^2 \rightarrow [0, 1] \quad (5)$$

denote an aggregation operator that combines the predictive score R_i and the contextual score Q_i into the final prioritization score

$$P_i = A(R_i, Q_i). \quad (6)$$

Throughout this work, aggregation operators are interpreted as integration policies governing the interaction between predictive evidence and contextual knowledge within the CADEMAs-ML architecture. **Rather than focusing on specific functional forms, the following definitions introduce a set of structural properties that characterize different aggregation semantics.**

Definition 1 (Boundedness). *An aggregation operator is bounded if*

$$0 \leq A(r, q) \leq 1, \quad \forall (r, q) \in [0, 1]^2. \quad (7)$$

Boundedness guarantees that the resulting prioritization score remains within the same normalized scale as the predictive and contextual assessments. **Since A maps $[0, 1]^2$ into $[0, 1]$, this property is automatically satisfied and is stated here only for completeness.**

Definition 2 (Monotonicity). *An aggregation operator is monotonic if*

$$r_1 \leq r_2, q_1 \leq q_2 \implies A(r_1, q_1) \leq A(r_2, q_2). \quad (8)$$

Monotonicity ensures that improving either predictive evidence or contextual satisfaction cannot decrease the final prioritization score.

Definition 3 (Continuity). *An aggregation operator is continuous if small perturbations in its inputs produce arbitrarily small changes in the aggregated score.*

Continuity is particularly relevant when predictive and contextual scores are estimated from noisy data, since it prevents abrupt changes in prioritization due to infinitesimal variations of the inputs.

Definition 4 (Zero Absorption). *An aggregation operator satisfies the zero-absorption property if*

$$A(r, 0) = 0, \quad \forall r \in [0, 1]. \quad (9)$$

Under zero absorption, a null contextual assessment completely suppresses the final prioritization score regardless of the predictive evidence available.

Definition 5 (Strict positivity on the interior). *An aggregation operator is strictly positive on the interior if*

$$A(r, q) > 0, \quad \forall (r, q) \in (0, 1]^2. \quad (10)$$

Zero absorption and strict positivity play complementary roles. The first forces a null score exactly when the context vetoes an alternative. The second guarantees that no other alternative is driven to zero. Taken together, they make the value $P_i = 0$ an unambiguous marker of a contextual veto, which is what allows veto preservation to be read directly from the aggregated scores.

Definition 6 (Contextual Veto Preservation). *An aggregation operator preserves contextual vetoes if*

$$Q_i = 0 \implies P_i = 0. \quad (11)$$

Within the CADEMAs-ML setting, contextual veto preservation means that contextual requirements cannot be overridden by predictive evidence. Observe that zero absorption is a sufficient condition for contextual veto preservation. **In fact, for $Q_i = 0$, zero absorption and contextual veto preservation are equivalent.**

Definition 7 (Compensability). *In this work, an aggregation operator is compensatory on an interior region if a decrease in one input can be locally offset by an increase in the other while preserving the aggregated score. For a differentiable operator, a sufficient local characterization is that both partial derivatives are strictly positive in that region.*

Thus, compensability denotes local substitutability rather than the ability to offset an arbitrary deficit. The unit-scale bounds limit every trade-off. It determines the extent to which strong predictive evidence may compensate for weak contextual support, or vice versa. At one extreme, fully compensatory operators permit the broadest within-scale trade-offs between prediction and context. At the opposite extreme, non-compensatory operators prevent such trade-offs and therefore behave as hard contextual constraints.

These properties provide a theoretical basis for distinguishing alternative integration semantics within CADEMAs-ML. In particular, zero absorption and compensability play a central role in determining whether contextual requirements behave as soft preferences, partial constraints, or strict vetoes. The following subsection introduces the specific aggregation operators considered in this work and analyzes how they instantiate these structural properties.

4.2. Aggregation Operators Considered

The CADEMAs framework does not prescribe a specific mechanism for combining predictive and contextual assessments [9]. Consequently, different aggregation operators can be incorporated without modifying either the predictive subsystem or the contextual subsystem.

To investigate how aggregation semantics affect prioritization behavior, policy compliance, and robustness, this work considers three representative operators that span different levels of compensability between predictive evidence and contextual requirements.

1. *Linear aggregation.* The linear operator corresponds to the integration mechanism adopted in existing CADEMAs-ML implementations [10]:

$$A_L(R_i, Q_i) = \lambda R_i + (1 - \lambda)Q_i, \quad \lambda \in [0, 1]. \quad (12)$$

The parameter λ controls the relative importance assigned to predictive and contextual information. For $\lambda \in (0, 1)$, this operator is compensatory: reductions in one component can be offset by increases in the other, subject to the bounds of the unit interval. The boundary cases $\lambda = 0$ and $\lambda = 1$ are degenerate and are not considered as compensatory regimes in the subsequent analysis.

2. *Weighted geometric aggregation.* An intermediate alternative is the weighted geometric mean:

$$A_G(R_i, Q_i) = R_i^\lambda Q_i^{1-\lambda}, \quad \lambda \in (0, 1). \quad (13)$$

Unlike the linear operator, the weighted geometric mean penalizes imbalanced evaluations and therefore reduces the degree of compensation between predictive and contextual information. Nevertheless, partial compensation remains possible whenever both inputs are strictly positive. Restricting the exponent to $\lambda \in (0, 1)$ keeps both powers strictly between 0 and 1, so the indeterminate form 0^0 never arises and $A_G(r, 0) = 0$ holds for every r . The two excluded endpoints behave differently: at $\lambda = 1$ the operator reduces to $A_G(R_i, Q_i) = R_i$ and loses the veto property, and at $\lambda = 0$ it reduces to Q_i . Both are therefore excluded from the analysis, and the implementation assigns $A_G = 0$ whenever either input equals zero.

3. *Minimum aggregation.* The third regime is the minimum operator

$$A_M(R_i, Q_i) = \min(R_i, Q_i). \quad (14)$$

This operator represents a non-compensatory integration regime in which the final prioritization score cannot exceed the weakest component. Consequently, predictive evidence cannot compensate for insufficient contextual support. Note that A_M carries no weighting parameter. Its output is invariant under changes of λ , so any comparison across integration regimes leaves the minimum operator unchanged by construction. This point matters when reading the λ -dependent results of Sections 5.2 and ??.

Table 2 summarizes the structural properties of the three operators according to the definitions introduced in the previous subsection.

Table 2. Structural properties of the aggregation operators considered in this work.

Property	Linear	Geometric	Minimum
Boundedness	Yes	Yes	Yes
Monotonicity	Yes	Yes	Yes
Continuity	Yes	Yes	Yes
Zero absorption	No [†]	Yes	Yes
Contextual veto preservation	No [†]	Yes	Yes
Strict positivity on $(0, 1]^2$	Yes	Yes	Yes
Compensability	Full [‡]	Partial	No

[†] For the linear operator, zero absorption and contextual veto preservation hold only in the degenerate case $\lambda = 0$.

[‡] Full compensability for the linear operator is understood for $\lambda \in (0, 1)$, with compensation bounded by the $[0, 1]$ scale.

All three operators are strictly positive on the interior, so they differ only in zero absorption. This isolates the property responsible for the contrasting policy behaviour reported in Section ??: the operators agree on which alternatives deserve a positive score and disagree only on whether a contextual veto can be overridden.

These operators were selected because they represent three distinct integration semantics within CADEMAs-ML: *full within-scale* compensation, partial compensation, and strict contextual constraint. The following subsections analyze the theoretical consequences of these semantics and their implications for prioritization behavior. *Here, full within-scale compensation refers to the linear operator with $\lambda \in (0, 1)$; it does not mean unbounded compensation.*

4.3. Conditions for Contextual Veto Preservation

Within CADEMAs-ML, contextual information may represent different types of requirements, including organizational priorities, regulatory constraints, safety conditions, or expert-defined restrictions. In some applications, a contextual assessment should not behave as a soft preference but rather as a hard constraint. This situation occurs when a complete contextual violation must prevent an alternative from being prioritized, regardless of its predictive evidence.

This behavior can be characterized through the zero-absorption property introduced in Section 4.1. An aggregation operator provides contextual veto preservation if

$$A(r, 0) = 0, \quad \forall r \in [0, 1]. \quad (15)$$

Therefore, the question of whether CADEMAs-ML can enforce contextual vetoes reduces to determining whether the selected aggregation operator satisfies zero absorption.

Theorem 1 (Contextual veto preservation). *Let $A : [0, 1]^2 \rightarrow [0, 1]$ be an aggregation operator used in CADEMAs-ML. If A satisfies zero absorption, then every decision case i with a null contextual assessment,*

$$Q_i = 0, \quad (16)$$

receives a null prioritization score,

$$P_i = A(R_i, Q_i) = 0, \quad (17)$$

independently of its predictive score R_i .

Proof. The result follows directly from the definition of zero absorption. Since $Q_i = 0$, it follows that $P_i = A(R_i, Q_i) = A(R_i, 0)$. Because A satisfies zero absorption, $A(R_i, 0) = 0$, for every $R_i \in [0, 1]$. Therefore, $P_i = 0$. Hence, a complete contextual violation completely suppresses the final prioritization score regardless of the predictive evidence. $\square$

Corollary 1 (Veto preservation of the considered operators). *Among the aggregation operators considered in this work, the minimum and weighted geometric operators preserve contextual vetoes, whereas the linear operator does not for $\lambda \in (0, 1]$, with the degenerate case $\lambda = 0$ being an exception.*

Proof. For the minimum operator, $A_M(r, 0) = \min(r, 0) = 0$. Similarly, for the weighted geometric operator, $A_G(r, 0) = r^\lambda 0^{1-\lambda} = 0$, for $\lambda \in (0, 1)$. In contrast, for $\lambda \in (0, 1]$, the linear operator yields $A_L(r, 0) = \lambda r$, which is greater than zero whenever $r > 0$ and $\lambda > 0$. Therefore, predictive evidence can override a complete contextual violation under linear aggregation. For the degenerate case $\lambda = 0$, $A_L(r, 0) = 0$, so the linear operator trivially preserves contextual vetoes. $\square$

Theorem 1 constrains individual prioritization scores. In a Top-K selection, however, the relevant question is whether these score-level guarantees are sufficient to exclude

vetoed alternatives from the selected set. The following corollary makes this connection explicit and identifies the additional population-level condition required.

Let

$$\mathcal{V} = \{i : Q_i = 0\}$$

denote the set of vetoed alternatives, and let $\mathcal{T}_K$ denote the set of alternatives selected among the K largest prioritization scores.

Corollary 2 (Exclusion of vetoed alternatives from the Top- K set). *Let A satisfy zero absorption and strict positivity on the interior (Definition 5). Consider a population of N alternatives, of which N_V are vetoed ($Q_i = 0$), while every remaining alternative satisfies $R_i > 0$ and $Q_i > 0$. If*

$$N - N_V \geq K,$$

then

$$\mathcal{T}_K \cap \mathcal{V} = \emptyset.$$

Proof. By zero absorption, every vetoed alternative receives a zero prioritization score,

$$P_i = A(R_i, 0) = 0, \quad i \in \mathcal{V}.$$

By strict positivity on the interior, every non-vetoed alternative satisfying $R_i > 0$ and $Q_i > 0$ receives a strictly positive score,

$$P_i = A(R_i, Q_i) > 0, \quad i \notin \mathcal{V}.$$

Since $N - N_V \geq K$, there are at least K non-vetoed alternatives with strictly positive prioritization scores. Consequently, the K largest scores are all strictly positive, whereas every vetoed alternative has score zero. Hence no vetoed alternative can belong to the Top- K set, and therefore

$$\mathcal{T}_K \cap \mathcal{V} = \emptyset.$$

□

The corollary shows that zero absorption provides a score-level guarantee that extends to Top- K selection when a sufficient number of admissible alternatives is available. Thus, zero absorption alone is not sufficient to characterize the selected set: a population-level supply condition, $N - N_V \geq K$, is also required. If fewer than K non-vetoed alternatives are available, some vetoed alternatives may necessarily remain in the selected set, even though the aggregation operator assigns them a zero prioritization score.

The previous result establishes that contextual veto preservation is not a consequence of the predictive model or the contextual knowledge representation, but rather an emergent property of the integration semantics. Consequently, two CADEMAs-ML systems with identical predictive and contextual components may exhibit fundamentally different policy behavior depending only on the selected aggregation operator.

This observation motivates the subsequent analysis of rank reversal conditions and compensation thresholds, which characterize how much contextual improvement is required to offset predictive advantages under different aggregation regimes.

4.4. Conditions for Rank Reversal under Compensation

The previous section established that contextual veto preservation depends on the existence of a zero-absorption property. However, many aggregation operators used in decision-support systems follow a different philosophy: they allow partial compensation

between the aggregated sources of evidence. In such cases, a sufficiently high value in one dimension may offset a lower value in another dimension.

This property is common in compensatory aggregation schemes, including weighted arithmetic means and several non-linear aggregation operators. Although compensation provides flexibility when predictive evidence and contextual information represent partially substitutable criteria, it also introduces the possibility that the final prioritization ranking differs from the ranking induced by predictive evidence alone.

To formalize this phenomenon, let $A : [0, 1]^2 \rightarrow [0, 1]$ be a continuous aggregation operator that is strictly increasing in both arguments. Given two alternatives i and j , assume that alternative i has stronger predictive evidence, but alternative j has better contextual alignment: $R_i > R_j$ and $Q_i < Q_j$. The final prioritization scores are then $P_i = A(R_i, Q_i)$ and $P_j = A(R_j, Q_j)$. A rank reversal occurs when $P_j > P_i$, meaning that the contextual advantage of alternative j compensates for its predictive disadvantage.

Theorem 2 (General rank reversal condition under compensation). *Let $A : [0, 1]^2 \rightarrow [0, 1]$ be a continuous aggregation operator strictly increasing in both arguments. For two alternatives i and j satisfying predictive dominance $R_i > R_j$, assume the operator provides sufficient compensatory capacity such that there exists a contextual assessment $q_{\max} \leq 1$ satisfying $A(R_j, q_{\max}) > A(R_i, Q_i)$.*

Under these conditions, a rank reversal in favour of alternative j occurs if and only if $Q_j > Q_i + \Phi_A(R_i, R_j, Q_i)$, where Φ_A is the minimum contextual improvement required to compensate for the predictive difference, implicitly defined by $A(R_j, Q_i + \Phi_A) = A(R_i, Q_i)$.

Proof. Fix $R_i > R_j$ and Q_i . Define the difference function $g(q) = A(R_j, q) - A(R_i, Q_i)$. Because A is continuous and strictly increasing in its second argument, $g(q)$ is also continuous and strictly increasing. At $q = Q_i$, $g(Q_i) = A(R_j, Q_i) - A(R_i, Q_i) < 0$ because A is strictly increasing in its first argument and $R_j < R_i$. By hypothesis, there exists $q_{\max}$ such that $g(q_{\max}) > 0$. Therefore, by the Intermediate Value Theorem, there exists a unique contextual value $q^* \in (Q_i, q_{\max})$ such that $g(q^*) = 0$, which implies $A(R_j, q^*) = A(R_i, Q_i)$. Defining the required threshold as $\Phi_A(R_i, R_j, Q_i) = q^* - Q_i$, the relation $P_j > P_i$ (or $g(Q_j) > 0$) holds if and only if $Q_j > q^*$, which is equivalent to $Q_j > Q_i + \Phi_A(R_i, R_j, Q_i)$. $\square$

Under its stated feasibility condition, the theorem shows that rank reversal need not be an artefact of a particular aggregation formula: it can arise when predictive and contextual information are allowed to compensate for one another. The specific threshold Φ_A depends on the operator and on the two alternatives being compared. The theorem does not imply that every compensatory operator, or every pair of alternatives, admits a reversal within the unit interval.

This distinction is particularly relevant in CADEMAs-ML because the two aggregated components do not represent equivalent evaluation criteria. The predictive score R_i summarizes evidence generated by machine learning models, whereas the contextual score Q_i represents compliance with organizational, regulatory, or domain-specific requirements. Therefore, the choice of aggregation operator implicitly defines the degree to which contextual requirements can modify or override data-driven recommendations.

4.5. Sensitivity to Predictive and Contextual Perturbations

The previous sections analyzed the structural properties of aggregation operators in terms of compensation and contextual veto preservation. However, in practical decision-support systems, both predictive scores and contextual assessments are subject to uncertainty. Predictive models may produce unstable estimates due to limited data, model

uncertainty, or distribution changes, whereas contextual assessments may vary because of incomplete knowledge, expert disagreement, or evolving organizational requirements.

Therefore, an additional relevant property of an aggregation mechanism is its sensitivity to perturbations in its input dimensions. In the context of CADEMAs-ML, this property determines how strongly changes in predictive evidence or contextual requirements affect the final prioritization score.

Let $A : [0, 1]^2 \rightarrow [0, 1]$ be an aggregation operator. Consider a perturbation ϵ applied to the contextual assessment while the predictive score remains fixed. The resulting variation in the prioritization score is $\Delta_Q(\epsilon) = |A(R, Q + \epsilon) - A(R, Q)|$. Similarly, a perturbation in the predictive assessment produces $\Delta_R(\epsilon) = |A(R + \epsilon, Q) - A(R, Q)|$. These quantities measure the local influence of each information source on the final decision score. Operators that strongly amplify small perturbations may lead to unstable rankings, whereas operators with bounded sensitivity provide greater robustness against uncertainty in predictive or contextual information.

Theorem 3 (Lipschitz sensitivity of aggregation operators). *Let $A : [0, 1]^2 \rightarrow [0, 1]$ be an aggregation operator continuously differentiable on an open neighbourhood containing the line segment between (R, Q) and $(R + \epsilon_R, Q + \epsilon_Q)$. Assume that both endpoints, and hence the whole segment by convexity, lie in $[0, 1]^2$. If there exist finite constants L_R and L_Q such that*

$$|\partial A / \partial R| \leq L_R \quad \text{and} \quad |\partial A / \partial Q| \leq L_Q,$$

then, for any perturbations ϵ_R and ϵ_Q ,

$$|A(R + \epsilon_R, Q + \epsilon_Q) - A(R, Q)| \leq L_R |\epsilon_R| + L_Q |\epsilon_Q|.$$

Proof. By the Mean Value Theorem, there exists a point (ξ_R, ξ_Q) between (R, Q) and $(R + \epsilon_R, Q + \epsilon_Q)$ such that

$$A(R + \epsilon_R, Q + \epsilon_Q) - A(R, Q) = (\partial A / \partial R)(\xi_R, \xi_Q) \epsilon_R + (\partial A / \partial Q)(\xi_R, \xi_Q) \epsilon_Q.$$

Taking absolute values and applying the derivative bounds yields

$$|A(R + \epsilon_R, Q + \epsilon_Q) - A(R, Q)| \leq L_R |\epsilon_R| + L_Q |\epsilon_Q|.$$

□

The theorem establishes a local robustness bound. The constants L_R and L_Q quantify the maximum amplification of perturbations on the particular region considered; they do not by themselves establish a global ordering of operators or the stability of a population-level Top- K decision. Those questions are examined empirically in Section ??.

For the linear aggregation operator, $A_L(R, Q) = \lambda R + (1 - \lambda)Q$, the sensitivity coefficients are constant: $L_R = \lambda$ and $L_Q = 1 - \lambda$. Therefore, the influence of predictive and contextual uncertainty is uniform throughout the decision space. This property explains the fully compensatory behaviour of the linear operator: a perturbation in one dimension has exactly the same marginal effect regardless of the current values of prediction and context.

For the weighted geometric operator, $A_G(R, Q) = R^\lambda Q^{1-\lambda}$, the local sensitivities are $\partial A_G / \partial R = \lambda R^{\lambda-1} Q^{1-\lambda}$ and $\partial A_G / \partial Q = (1 - \lambda) R^\lambda Q^{-\lambda}$. Unlike the linear operator, these sensitivities depend on the current decision state. In particular, $\partial A_G / \partial Q$ becomes unbounded as $Q \rightarrow 0^+$ when $R > 0$. Thus, finite Lipschitz constants for A_G are available only on regions bounded away from the coordinate axes. Near a contextual veto, a small positive

observational error can therefore have a large local effect on the aggregated score. This fact is consistent with, but does not by itself determine, a false-negative-veto mechanism under imperfect contextual observation.

For the minimum operator, $A_M(R, Q) = \min(R, Q)$, the final score is completely determined by the limiting component. Consequently, the operator exhibits zero compensation: improving the non-limiting input has no effect until it becomes the active constraint. This property provides maximal preservation of contextual restrictions but may reduce responsiveness when contextual assessments are uncertain. Because A_M is not differentiable on $R = Q$, it is not covered by Theorem 3 there; the following corollary captures one of its piecewise-stable regimes.

Corollary 3 (Minimum operator stability under inactive-component perturbations). *Consider a decision case evaluated under the minimum operator $A_M(R, Q) = \min(R, Q)$. If the predictive score acts as the strict limiting component ($R < Q$), the operator completely absorbs any contextual perturbation ε as long as the ordinal relationship is maintained (i.e., $R \leq Q + \varepsilon$). Under these conditions, the sensitivity is zero:*

$$|A_M(R, Q + \varepsilon) - A_M(R, Q)| = |R - R| = 0.$$

Proof. Let the initial state of the decision case be defined by R and Q . By hypothesis, the predictive score is the strict limiting component, meaning $R < Q$. By the definition of the minimum operator, the initial aggregated score is:

$$A_M(R, Q) = \min(R, Q) = R.$$

Now, consider a perturbed contextual assessment $Q' = Q + \varepsilon$. By hypothesis, the perturbation ε (whether positive or negative) is bounded so that the ordinal relationship is maintained, that is, $R \leq Q + \varepsilon$. Evaluating the minimum operator under this perturbed state yields:

$$A_M(R, Q + \varepsilon) = \min(R, Q + \varepsilon) = R.$$

Calculating the absolute difference between the perturbed and initial states gives the sensitivity:

$$|A_M(R, Q + \varepsilon) - A_M(R, Q)| = |R - R| = 0.$$

Thus, the perturbation is completely absorbed, and the final prioritization score remains unaffected. $\square$

This result illustrates the asymmetric robustness of non-compensatory operators. Perturbations of the non-limiting component are absorbed entirely, whereas perturbations of the active constraint pass through to the aggregated score. The transmission is never amplified: since $|\min(R, Q + \varepsilon) - \min(R, Q)| \leq |\varepsilon|$ for all R, Q, ε , the minimum operator is globally 1-Lipschitz in each argument, and its sensitivity therefore alternates between 0 and at most $|\varepsilon|$. The contrast with A_G is worth noting: the geometric operator is smooth but its contextual sensitivity grows without bound as $Q \rightarrow 0^+$, whereas A_M is non-smooth yet uniformly bounded. Smoothness and robustness are thus not equivalent here. These are score-level bounds, however, and they do not by themselves determine how stable the induced ranking is; that question is addressed empirically in Section ??.

These analytical differences motivate the robustness experiments conducted in the following sections. Rather than evaluating aggregation operators only according to final ranking performance, the experiments examine how their semantic properties influence stability under perturbations of predictive evidence and contextual assessments.

5. Simulation Results

The theoretical analysis establishes how the aggregation operators differ in terms of compensation, veto preservation, and sensitivity to perturbations. The simulation study complements these score-level results by examining their consequences when the aggregated scores are used to prioritize a population of competing alternatives. The experiments therefore focus on two aspects that cannot be determined from the operator definitions alone: how the resulting prioritization changes as the relative weight of predictive and contextual evidence varies, and how stable the induced rankings and Top- K selections remain under perturbations of the underlying scores.

The experimental design described below uses controlled Monte Carlo populations to isolate these effects while keeping the decision architecture fixed. The results are organized accordingly: first, we examine the λ -sensitivity of the three aggregation semantics, with particular attention to the compensatory behavior of the linear and geometric operators; second, we evaluate the stability of the induced rankings and Top- K selections under perturbations of the predictive score R . This latter analysis deliberately treats R as the uncertain input, reflecting the predictive nature of the ADM outputs in the CADEMAs-ML setting. Although contextual assessments Q could likewise be uncertain in other implementations, a separate Q -noise experiment is unnecessary here: the aggregation operators exhibit the corresponding argument-wise properties derived theoretically, so the conclusions obtained for perturbations of R extend directly to perturbations of Q under the analogous conditions. Throughout the analysis, the empirical findings are therefore interpreted in relation to the theoretical properties established in Section 4.

5.1. Experimental Design

The simulation study uses controlled Monte Carlo populations to isolate the effects of aggregation semantics while keeping the underlying decision architecture fixed. Each population contains $N = 1000$ alternatives, with each alternative i characterized by a predictive score $R_i \in [0, 1]$ and a contextual assessment $Q_i \in [0, 1]$, representing the outputs of the decision and contextual pathways, respectively. The same population is evaluated under the linear, weighted geometric, and minimum aggregation operators,

$$P_i = A(R_i, Q_i; \lambda),$$

which determines the resulting prioritization and Top- K selection. Unless otherwise stated, $K = 100$ and 20% of the population is assigned a contextual veto ($Q_i = 0$). Vetoed alternatives are generated with comparatively high predictive scores, creating explicit conflicts between predictive evidence and contextual admissibility. The remaining scores are sampled from bounded distributions designed to provide both aligned and conflicting combinations of R_i and Q_i , including weak-context cases with $0 < Q_i \leq 0.25$.

The first analysis performs a dense sweep of $\lambda \in [0, 1]$ to characterize how the relative contribution of predictive and contextual evidence modifies the resulting prioritization. The linear and weighted geometric operators are evaluated throughout this sweep, whereas the minimum operator is independent of λ and therefore provides a natural reference. For the complementary comparison of prioritization stability, we use seven representative aggregation configurations: $A_L^{0.10}$, $A_L^{0.50}$, $A_L^{0.90}$, $A_G^{0.10}$, $A_G^{0.50}$, $A_G^{0.90}$, and A_M . The three λ values represent low, balanced, and high relative weight on predictive evidence, respectively, while avoiding an unnecessarily large number of pairwise comparisons.

All experiments use 1000 Monte Carlo replications with registered seeds 42–1041. The evaluation considers three complementary aspects of the resulting prioritization: contextual

policy compliance, predictive utility, and decision stability. Let $\mathcal{T}_K$ denote the selected Top- K set and let

$$\mathcal{V} = \{i : Q_i = 0\}$$

denote the set of vetoed alternatives. Policy compliance is measured by the policy violation rate

$$V = \frac{|\mathcal{T}_K \cap \mathcal{V}|}{K},$$

which represents the proportion of selected alternatives that violate a contextual veto. The metric counts only strict vetoes ($Q_i = 0$) and does not treat weak non-veto cases ($0 < Q_i \leq 0.25$) as violations. Thus, $V = 0$ corresponds to complete preservation of the contextual vetoes, whereas larger values indicate greater policy non-compliance. Because K is fixed and positive, V is well defined for every experimental condition.

Predictive utility is characterized by the mean original predictive score of the selected alternatives,

$$\bar{R} = \frac{1}{K} \sum_{i \in \mathcal{T}_K} R_i.$$

The average is computed over the complete Top- K set, including vetoed alternatives when present. This keeps the denominator fixed and allows $\bar{R}$ to be interpreted alongside V : the former describes the predictive evidence represented in the selected set, whereas the latter describes its contextual compliance. In the perturbation experiments, $\bar{R}$ is computed from the original, unperturbed R_i values so that the metric measures the predictive quality of the selected alternatives rather than the magnitude of the injected noise.

The second analysis evaluates robustness to uncertainty in the predictive pathway by perturbing the predictive score according to

$$R'_i = \text{clip}(R_i + \epsilon_i, 0, 1), \quad \epsilon_i \sim \mathcal{N}(0, \sigma_R^2), \quad (18)$$

and comparing the resulting prioritization with the corresponding noiseless reference. We focus the uncertainty analysis on R because it represents the output of an ADM or predictive model and is therefore naturally subject to estimation error. No separate Q -noise experiment is required: the corresponding score-level sensitivity properties extend to Q by the symmetry of the aggregation operators. The clipping operation is retained because both scores are bounded to $[0, 1]$.

Decision stability is assessed from two complementary perspectives. For a reference aggregated-score vector $P^{(1)}$ and a perturbed or alternative vector $P^{(2)}$, Kendall's τ_b measures agreement between their global rankings, with the tie correction inherent to the statistic. In contrast, the Jaccard similarity measures agreement between the corresponding Top- K sets,

$$J = \frac{|\mathcal{T}_K^{(1)} \cap \mathcal{T}_K^{(2)}|}{|\mathcal{T}_K^{(1)} \cup \mathcal{T}_K^{(2)}|}.$$

Kendall's τ_b therefore captures changes distributed across the complete ranking, whereas J focuses directly on the stability of the alternatives that would actually be prioritized. For Top- K selection, ties in P are resolved using the predictive score available in the corresponding condition as a secondary criterion and the alternative identifier as a final deterministic criterion, ensuring reproducible selections.

Monte Carlo means are reported together with 95% percentile intervals, which describe the dispersion of outcomes across independently generated populations rather than the sampling uncertainty of the estimated mean. The simulation study is therefore intended to characterize the population-level consequences of the analytically established aggregation properties, rather than to reproduce their proofs.

5.2. Policy Compliance under Contextual Vetoes

The first experiment examines the population-level consequences of the contextual veto property established in Section 4.3. It addresses RQ2 by asking when the aggregation semantics preserve a contextual exclusion and when predictive evidence can compensate for it. The experiment therefore focuses on the interaction between aggregation and prioritization rather than on overall predictive performance.

We construct a controlled stress population in which 20% of the $N = 1000$ alternatives are contextual vetoes, represented by $Q_i = 0$. These alternatives are deliberately assigned comparatively high predictive scores, so that prediction and contextual admissibility are placed in direct conflict. The remaining alternatives provide both aligned and conflicting combinations of R_i and Q_i . In particular, the (R, Q) plane is partitioned by $R = 0.5$ and $Q = 0.5$ to distinguish cases in which predictive and contextual evidence broadly agree from those in which they disagree, including contextually weak cases with high predictive scores (Q -weak) and predictively weak cases with high contextual scores (R -weak). The prioritization selects the $K = 100$ highest aggregated scores. Since $N - N_V = 800 \geq K$, the supply condition of Corollary 2 is satisfied.

Figure 1 illustrates the resulting behaviour. Panel (a) shows the structure of the synthetic population in the (R, Q) plane. Panels (b)–(d) use the same population from one representative Monte Carlo replication to show the distributions of the aggregated score P_i for the three operators at $\lambda = 0.10, 0.50$, and 0.90 , respectively. The dashed horizontal line in each distribution is the Top- K cutoff for that configuration, while the overlaid markers identify the corresponding vetoed alternatives. These panels should be read together: panel (a) identifies where the potentially conflicting cases originate, whereas panels (b)–(d) show how the aggregation semantics translate those conflicts into the score used for prioritization. Panel (e) then moves from this illustrative population to the Monte Carlo experiment, reporting the mean policy violation rate over the dense $\lambda \in [0, 1]$ sweep.

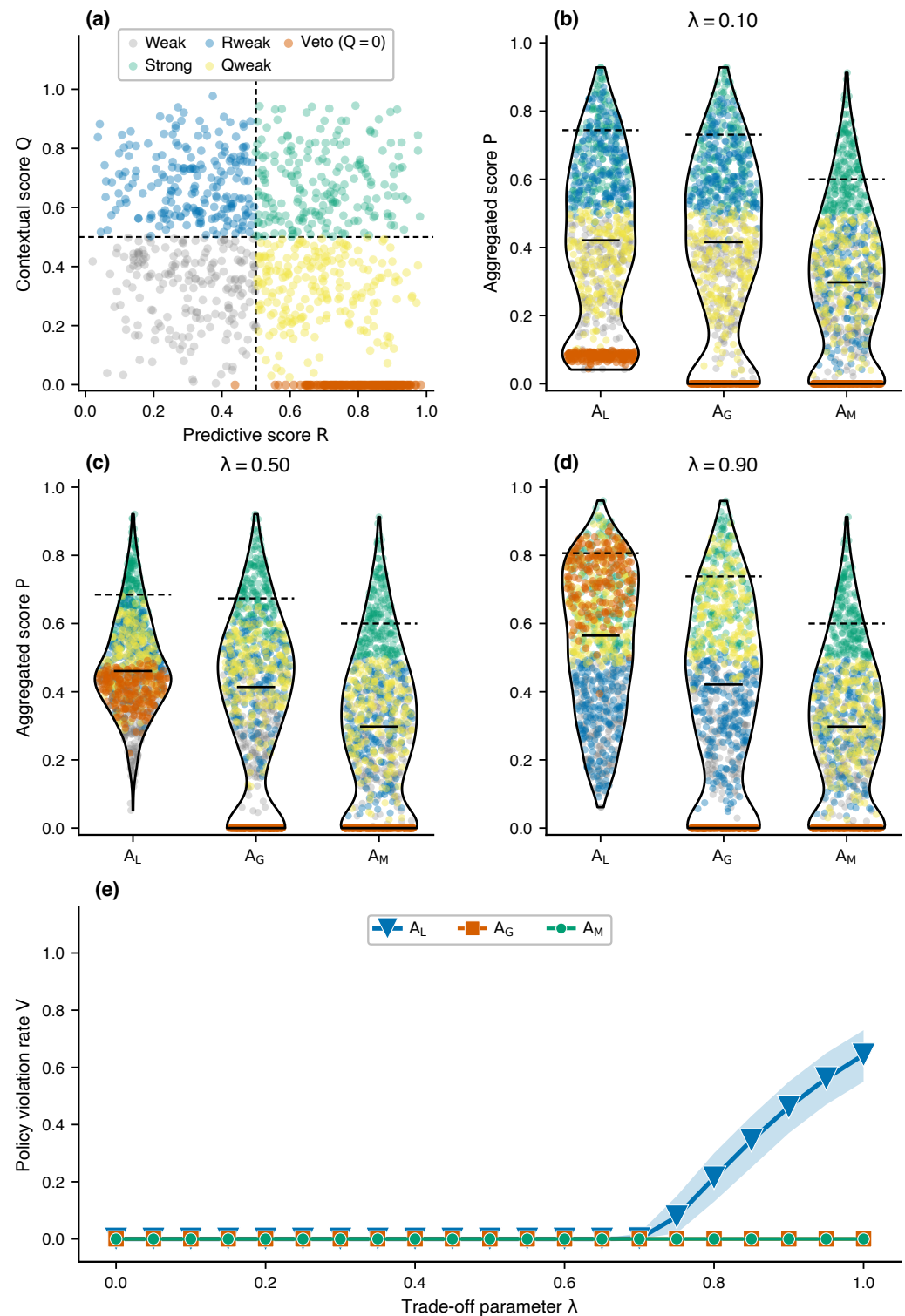


Figure 1. Policy compliance under contextual vetoes ($N = 1000$, $K = 100$). (a) Illustrative population in the (R, Q) plane, including contextually vetoed cases ($Q_i = 0$) and non-veto cases with different predictive-contextual profiles. (b–d) Aggregated-score distributions for A_L , A_G , and A_M at $\lambda = 0.10$, 0.50 , and 0.90 , respectively, for the same population. The dashed line indicates the Top-K cutoff and the overlaid points identify vetoed cases. (e) Monte Carlo mean policy violation rate over $\lambda \in [0, 1]$ across 1000 replications; shaded bands indicate 95% percentile intervals.

At low predictive weight, the three operators produce the same qualitative compliance outcome. The distinction becomes visible as λ increases. For the linear operator, vetoed alternatives with $Q_i = 0$ and $R_i > 0$ receive positive aggregated scores. Consequently,

their scores move upwards through the distribution as predictive evidence receives greater weight. In the representative population, this movement can be seen directly in panels (b)–(d): the veto markers for A_L progressively approach and eventually cross the Top- K cutoff. The corresponding population-level effect appears in panel (e). The mean violation rate remains zero at low λ , becomes positive at approximately $\lambda = 0.70$, and reaches $V = 0.462$ at $\lambda = 0.90$ (95% percentile interval $[0.370, 0.550]$). The precise onset is a property of the simulated population rather than a universal critical value of the linear operator.

The behaviour of the weighted geometric and minimum operators is different for a structural reason. Both are zero-absorbing, so $Q_i = 0$ implies $P_i = 0$ regardless of the value of R_i or of λ . This is visible in panels (b)–(d), where the vetoed alternatives remain on the zero-score boundary even as the distribution of non-vetoed alternatives changes. Because at least K non-vetoed alternatives remain and satisfy the positivity condition required by Corollary 2, these zero-valued vetoes cannot enter the Top- K set. Hence the observed $V = 0$ across the entire λ sweep is a deterministic consequence of the theoretical conditions, rather than evidence of an empirical performance advantage.

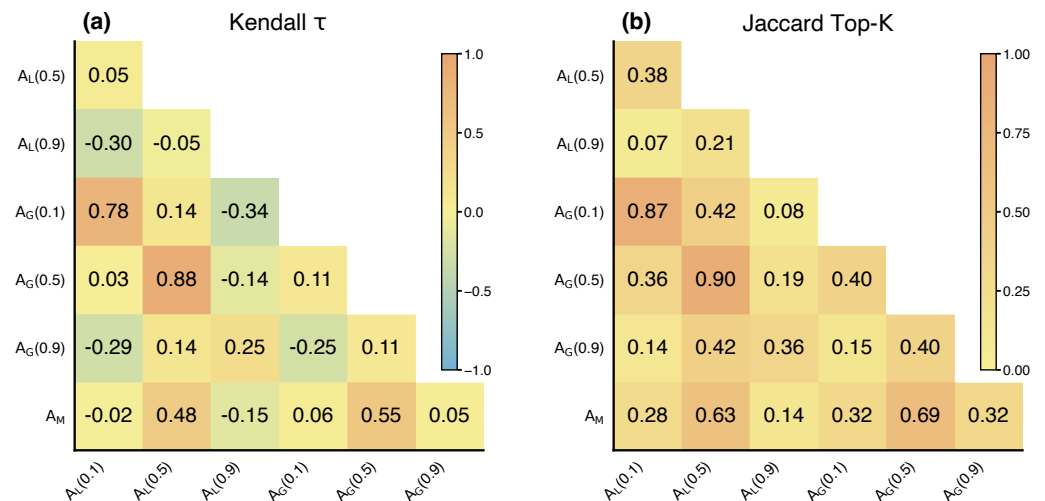


Figure 2. Pairwise agreement among the seven aggregation configurations considered in Experiment 01 ($N = 1000$, $K = 100$, 1000 Monte Carlo replications). The configurations are A_L and A_G at $\lambda \in \{0.10, 0.50, 0.90\}$ and A_M . Each cell reports the Monte Carlo mean of the corresponding pairwise statistic across replications. (a) Kendall's τ_b between the aggregated-score rankings. (b) Jaccard similarity between the corresponding Top- K sets. Higher values indicate greater agreement; the diagonal is omitted because self-comparisons are identically equal to one.

The experiment also reveals that the aggregation semantics affect more than the treatment of explicit vetoes. Figure 2 compares the resulting prioritizations across seven representative configurations: A_L and A_G at $\lambda \in \{0.10, 0.50, 0.90\}$, together with A_M . Kendall's τ_b measures agreement between their induced score rankings, whereas Jaccard similarity measures the overlap between the corresponding Top- K sets. Thus, the two statistics distinguish changes in the overall ordering from changes in the actual alternatives selected for intervention.

The pairwise results show that aggregation semantics can produce similar decisions when the predictive and contextual contributions are balanced, but the agreement can decrease substantially when a highly predictive weight is combined with a compensatory operator. For example, $A_L(0.50)$ and $A_G(0.50)$ retain a relatively high mean rank agreement ($\tau_b > 0.88$), whereas the comparison between $A_L(0.90)$ and $A_G(0.10)$ yields mean $\tau_b < 0$ and a Jaccard overlap below 0.15. The latter case illustrates a stronger form of semantic divergence: the two systems can assign substantially different priorities to the

same alternatives and consequently select largely different Top-K sets. This divergence should not be interpreted as evidence that one operator is generally superior. Rather, it shows that the choice of aggregation semantics is itself a consequential modelling decision in CADEMAs-ML.

These observations are consistent with Theorem 1 and Corollary 2. The theorem establishes the score-level consequence of zero absorption, while the corollary identifies the additional population-level condition under which this property becomes Top-K exclusion. The simulation shows the complementary case for the linear operator: because it is compensatory and does not satisfy zero absorption, the theory provides no exclusion guarantee, and the actual population distribution determines when vetoed alternatives become competitive. The dense sweep therefore quantifies a population-level effect that cannot be inferred from the operator definition alone.

Overall, the experiment shows two distinct consequences of aggregation semantics. Zero-absorbing operators preserve contextual vetoes under the conditions of Corollary 2, whereas the compensatory linear operator can trade contextual admissibility against predictive evidence as λ increases. At the same time, the Kendall and Jaccard results demonstrate that these semantic differences propagate to the final prioritization itself. The aggregation operator is therefore not merely a technical implementation detail: it can determine both whether excluded alternatives remain excluded and which alternatives ultimately constitute the Top-K decision set.

5.3. Sensitivity Analysis

The second experiment examines how uncertainty in the predictive pathway propagates to the prioritization produced by CADEMAs-ML, addressing RQ3 and RQ4. We focus on perturbations of the predictive score because R represents the output of an ADM or predictive model and can therefore be affected by estimation error. For the three aggregation operators considered here, the same analysis applies to contextual perturbations by exchanging R and Q and, for the linear and geometric operators, replacing λ by $1 - \lambda$. We therefore use noise on R as the representative uncertainty channel rather than duplicating an analytically symmetric experiment on Q .

For each of the 1000 Monte Carlo populations, the predictive scores are perturbed according to Eq. (18), with $\sigma_R \in \{0.05, 0.10, 0.20\}$. The contextual scores remain unchanged, including the 20% of cases with $Q_i = 0$. We evaluate the seven ranking configurations used throughout the experiments: A_L and A_G at $\lambda \in \{0.10, 0.50, 0.90\}$, together with A_M . These values provide low, intermediate, and high predictive weighting for the two λ -dependent operators, while A_M serves as the λ -independent reference. For each configuration, we record the policy violation rate V and compare the noisy ranking with the clean ranking obtained from the same population before perturbation, using Kendall's τ_b and the Top-K Jaccard similarity. We also compare the seven configurations with one another under the same noisy realization, which allows us to distinguish loss of stability relative to the clean ranking from changes in the agreement between aggregation semantics.

Figure 3 first illustrates how the perturbation changes the underlying population. Panels (a)–(c) show the same Monte Carlo population at increasing noise levels. Since Q is fixed, the perturbation acts horizontally in the (R, Q) plane. The vetoed cases therefore remain on the $Q = 0$ boundary, whereas non-vetoed alternatives are increasingly dispersed in the predictive dimension. At the largest noise level, the clipping operation also becomes visible through the accumulation of observations at the boundaries of the admissible range. This is important for interpreting the subsequent ranking results: the perturbation is not equivalent to an unconstrained Gaussian displacement, but to noisy scores projected back onto $[0, 1]$.

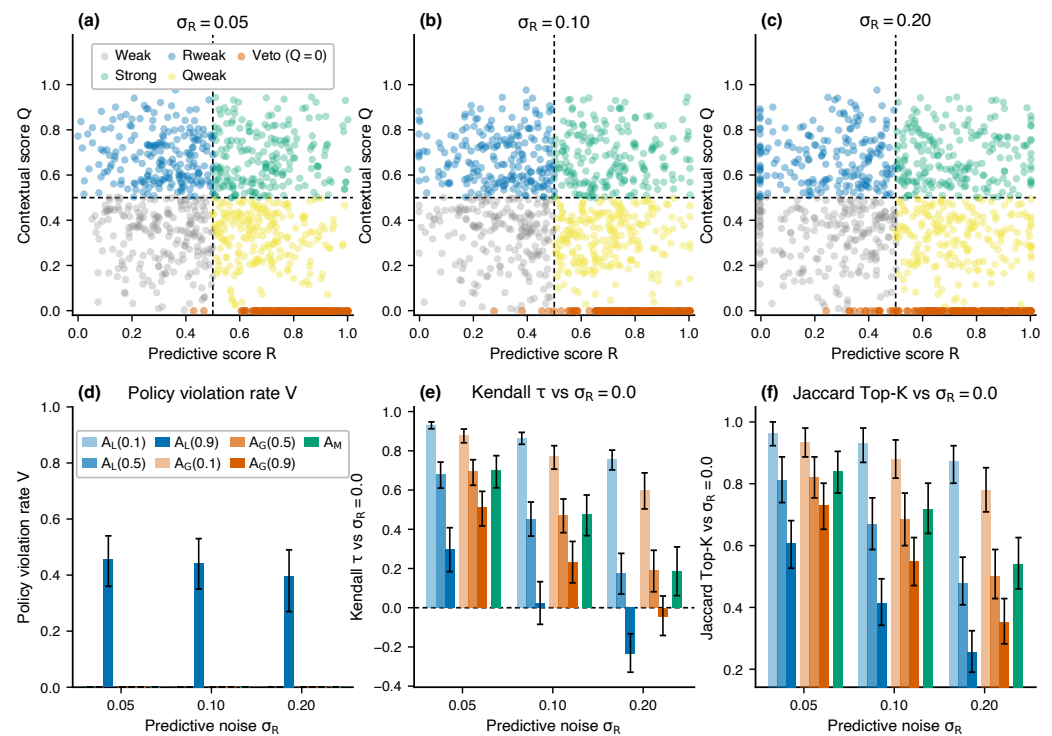


Figure 3. Sensitivity to predictive-score perturbations. Panels (a)–(c) show one illustrative population at increasing noise levels. Panels (d)–(f) summarize the corresponding Monte Carlo results for the seven ranking configurations.

The population-level consequences are shown in panels (d)–(f). The first pattern concerns policy compliance. The zero-absorbing operators A_G and A_M retain $V = 0$ across the tested noise levels, whereas noise does not provide any additional protection for the compensatory linear operator. In particular, the high-predictive configuration $A_L(0.90)$ continues to exhibit substantial violations at the larger noise levels. This is consistent with Theorem 1: perturbing R does not alter the fact that A_G and A_M map a true contextual veto $Q_i = 0$ to zero. Since the population satisfies the supply condition of Corollary 2, this score-level property continues to exclude vetoed alternatives from the Top- K set.

The second pattern is the loss of agreement with the clean ranking. At $\sigma_R = 0.05$, the perturbation produces a moderate degradation for most configurations, but the effect becomes much stronger as the noise increases. The degradation is also operator-dependent: configurations that give greater weight to R are generally more sensitive, particularly for the linear operator. At $\sigma_R = 0.20$, for example, the mean Kendall correlation for $A_L(0.90)$ falls to approximately -0.24 , while its Top- K Jaccard similarity is only about 0.25. Thus, at sufficiently large perturbations, the noisy prioritization can differ substantially from the clean one, both in the global ordering and in the actual set of selected alternatives. The stronger decrease in Jaccard than in Kendall's τ_b indicates that the instability is particularly consequential near the Top- K boundary.

The comparison between operators under the same noisy realizations is shown in Figure 4. The heatmaps provide a complementary view of the previous analysis: rather than asking whether each configuration remains close to its own clean ranking, they ask whether different aggregation semantics continue to select similar alternatives when exposed to the same uncertain predictive scores. Agreement generally decreases as σ_R increases, but the reduction is not uniform across configurations. In particular, configurations with different λ values can diverge substantially, while configurations with similar integration semantics remain more closely aligned. The result therefore confirms that uncertainty does not

erase the effect of aggregation semantics; instead, it interacts with the way each operator translates the perturbed evidence into prioritization scores.

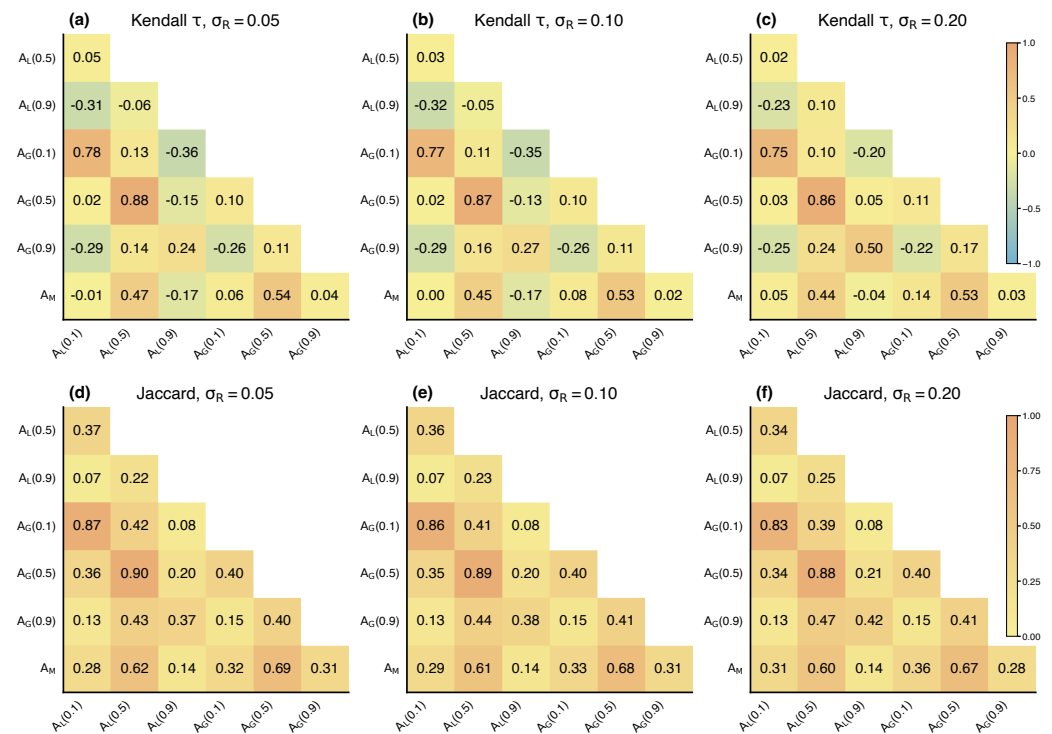


Figure 4. Pairwise agreement among the seven ranking configurations under shared predictive-score perturbations. The heatmaps report mean Kendall τ_b and Top-K Jaccard similarity across Monte Carlo replications.

Taken together, the experiment shows two distinct effects of predictive uncertainty. It can alter the prioritization produced by all operators, especially when predictive evidence receives high weight, but it does not remove the structural distinction established by the theory. Zero absorption protects the veto mechanism of A_G and A_M , whereas the linear operator remains compensatory and can therefore propagate high predictive scores into contextually excluded alternatives. The sensitivity analysis consequently complements the deterministic results of the previous experiment: uncertainty changes the stability of the resulting ranking, but it does not change the underlying aggregation semantics that determine whether contextual vetoes are structurally preserved.

6. Employee Attrition Case Study

6.1. Original CADEMAs-ML Application

6.2. Experimental Setup

6.3. Results and Interpretation

7. Discussion

7.1. Implications for Cooperative Automated Decision Making

7.2. Relation to Fuzzy MCDM and Aggregation Theory

7.3. Limitations

8. Conclusions and Future Work

Author Contributions: To be completed.

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